

Optimization on $SO(3)$ and Quaternions

Optimization and Learning for Robot Control

RCI Lab

Kyung Hee University

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Where We Are

We want to run:

- Newton
- Gauss-Newton
- DDP
- SQP

on states that include rotations.

Rotations live on:

$$SO(3)$$

This is a manifold — not a vector space.

Geometry of Unit Quaternions

Quaternion:

$$q = \begin{bmatrix} q_0 \\ \mathbf{q} \end{bmatrix}$$

Constraint:

$$\|q\| = 1$$

Thus:

$$q \in S^3 \subset \mathbb{R}^4$$

Tangent space at identity:

$$T_1 S^3 \cong \mathbb{R}^3$$

Derivatives live in the tangent space.

Infinitesimal Rotation

Small axis-angle:

$$\delta\theta \in \mathbb{R}^3$$

Associated quaternion:

$$\delta q \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix}$$

This lies in tangent space.

Multiplicative Update

Rotations compose by multiplication:

$$q_{new} = \delta q \otimes q$$

NOT:

$$q + \delta q$$

This is the key difference from Euclidean optimization.

Suppose:

$$f(q) : SO(3) \rightarrow \mathbb{R}$$

We compute derivative w.r.t. perturbation:

$$\left. \frac{d}{d\epsilon} f(\delta q(\epsilon) \otimes q) \right|_{\epsilon=0}$$

Gradient lives in:

$$\mathbb{R}^3$$

not $\mathbb{R}^4$.

Quaternion Jacobian Structure

Let:

$$f(q) \in \mathbb{R}^m$$

Linearization:

$$f(q \oplus \delta\theta) \approx f(q) + J_q \delta\theta$$

where:

$\oplus$ denotes multiplicative update

This Jacobian maps:

$$\mathbb{R}^3 \rightarrow \mathbb{R}^m$$

Hessian on $SO(3)$

Second order expansion:

$$f(q \oplus \delta\theta) \approx f(q) + g^T \delta\theta + \frac{1}{2} \delta\theta^T H \delta\theta$$

Now Newton step:

$$H\delta\theta = -g$$

Update:

$$q \leftarrow \text{Exp}(\delta\theta) \otimes q$$

Wahba's Problem

Given known world vectors:

$$x_i$$

Measured body vectors:

$$y_i$$

Find rotation:

$$\min_q \sum_{i=1}^N \|x_i - R(q)y_i\|^2$$

Residual Formulation

Define residual:

$$r_i(q) = x_i - R(q)y_i$$

Cost:

$$J(q) = \sum_i r_i^T r_i$$

Gauss-Newton approximation:

$$H \approx J_q^T J_q$$

$$g = J_q^T r$$

Gauss-Newton on $SO(3)$

Compute:

$$\delta\theta = -(J_q^T J_q)^{-1} J_q^T r$$

Multiplicative update:

$$q \leftarrow \text{Exp}(\delta\theta) \otimes q$$

Repeat until convergence.

Why This Matters

Now we can:

- Run Newton on rotations
- Run DDP with quaternion states
- Run SQP with rotational constraints
- Do pose estimation properly

All without singularities.